

SUPPLEMENTARY PROBLEM: ASYMPTOTIC NOTATIONS

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Problem Statement

For each of the following pairs of functions, write down the strictest possible asymptotic relation between $f(n)$ and $g(n)$; i.e., if $f(n) = x(g(n))$, where $x \in \{o, \Theta, \omega\}$, write down x under the “Relation” column in the corresponding row (an example is provided on the 0th row).

#	$f(n)$	$g(n)$	Relation
0	$n^2 + 1$	$2n^2$	Θ
1	$2n^2$	$\frac{n^3}{3}$	
2	n^n	2^n	
3	$(0.1)^n$	$(0.2)^n$	
4	$\log\left(\frac{n}{1000}\right)$	$\log(n^{1000})$	
5	$1 + \frac{1}{n}$	1	
6	2^{-n}	$\frac{1}{n}$	
7	$(n + 1000)^2$	$(n - 1000)^3$	
8	$\sum_{i=1}^n i$	$\sum_{i=\lfloor n/10 \rfloor}^n i$	

Solutions

#	Relation
0	Θ
1	o
2	ω
3	o
4	Θ
5	Θ
6	o
7	o
8	Θ

Solution Explanations

0. Choose $c_1 = 1, c_2 = 10$ to achieve the following inequality that will hold for $n \geq 1$.

$$1 * n^2 \leq n^2 + n \leq 10 * n^2$$

1. Evaluate the following limit. By definition, $f(n) \in o(g(n))$, or more intuitively $g(n)$ is strictly greater than $f(n)$.

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \lim_{n \rightarrow \infty} \frac{2n^2}{\frac{n^3}{3}} = \lim_{n \rightarrow \infty} \frac{6n^2}{n^3} = 0$$

2. Evaluate the following limit. By this definition, $f(n) \in \omega(g(n))$, or in other words $g(n)$ is strictly less than $f(n)$.

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \lim_{n \rightarrow \infty} \frac{n^n}{2^n} = \lim_{n \rightarrow \infty} \left(\frac{n}{2}\right)^n = \infty$$

3. Evaluate the following limit. By this definition, $f(n) \in o(g(n))$, or in other words $g(n)$ is strictly greater than $f(n)$.

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \lim_{n \rightarrow \infty} \frac{(0.1)^n}{(0.2)^n} = \lim_{n \rightarrow \infty} \left(\frac{1}{2}\right)^n = 0$$

4. Choose $c_1 = 1/10000, c_2 = 1, n_0 = 2155$. Then for sufficiently large $n \geq n_0$, the following inequality holds, therefore $f(n) \in \Theta(g(n))$.

$$\frac{1}{10} \log(n) \leq \log(n) - \log(1000) \leq 1000 \log(n)$$

5. Choose $c_1 = 1, c_2 = 2$ to achieve the following inequality for all positive n , therefore $f(n) \in \Theta(g(n))$.

$$1 \leq 1 + \frac{1}{n} \leq 2$$

6. Evaluate the following limit. By definition, $f(n) \in o(g(n))$

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \lim_{n \rightarrow \infty} \frac{\frac{1}{2^n}}{\frac{1}{n}} = \lim_{n \rightarrow \infty} \frac{n}{2^n} = 0$$

7. Evaluate the following limit. By this definition, $f(n) \in o(g(n))$.

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \lim_{n \rightarrow \infty} \frac{n^2 + 2000n + 1000000}{n^3 - 3000n^2 + 1000^2 * 3n - 1000^3} = 0$$

8. For convenience, rewrite $\sum_{i=\lfloor n/10 \rfloor}^n i = [\sum_{i=1}^n i] - [\sum_{i=1}^{\lfloor n/10 \rfloor} i]$.

Knowing that $\sum_{i=1}^n i = \frac{n^2+n}{2}$ allows us to calculate $g(n) = \frac{9}{10}(\frac{n^2+n}{2})$. Therefore, choose $c_1 = 1, and c_2 = 2$ to achieve the following inequality

$$\frac{9}{10} \left(\frac{n^2 + n}{2} \right) \leq \frac{n^2 + n}{2} \leq \frac{9}{5} \left(\frac{n^2 + n}{2} \right)$$